

Smart Home Health Monitor

Project Plan Preparation (PPP)

Mausam Shrestha

CSC 494 - IoT

Spring 2026

The Problem

Current Air Purifiers Have Critical Flaws

1. **Location Bias:** Sensors measure air inside the purifier (already cleaned), not where you breathe
2. **Limited Monitoring:** Only PM2.5, missing CO2 and VOCs
3. **Poor Accuracy:** Cheap infrared sensors with 30-40% error rates

Result: People think their air is clean when it's not

Why This Matters

Research-Backed Health Impact

- **Sleep Quality:** High PM2.5/NO2 increases sleep issues by 50-60%
- **Cognitive Function:** CO2 above 1000 ppm significantly reduces productivity
- **Market Need:** 73 million baby boomers need better home air quality
- **User Complaints:** 10,000+ mentions of device disconnections and inaccuracy

People spend 90% of time indoors - air quality directly impacts health

The Solution

Smart Monitor at Breathing Zone Height

Place sensors where you breathe (3-6 ft), not at purifier

Monitor 5 parameters:

- CO2 (cognitive function indicator)
- PM2.5 (particulate matter)
- VOC (volatile organic compounds)
- Temperature
- Humidity

Control any purifier via smart plug based on accurate readings

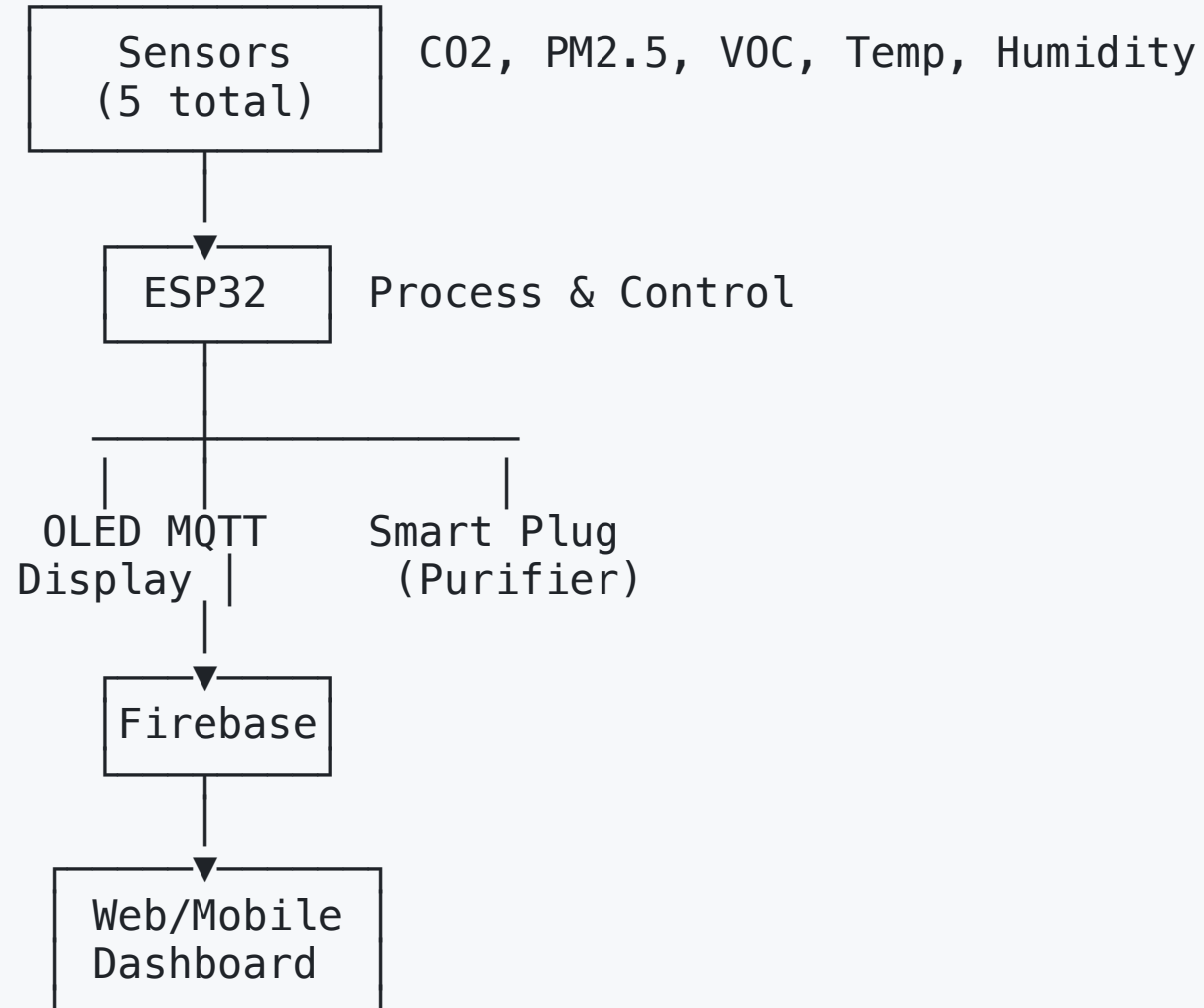
Unique Value Proposition

Why Build This vs Buy Commercial?

Feature	Commercial	Our Solution
Sensor Location	Inside purifier	Breathing zone
Parameters	PM2.5 only	5 parameters
Compatibility	Brand-locked	Any purifier
Data Access	Closed	Open/exportable
Cost (3 rooms)	\$600-900	\$300

Plus: Learning opportunity and portfolio piece

System Architecture



Hardware Components

Total Cost: \$88

- **ESP32 DevKitC** (\$8) - WiFi microcontroller
- **MH-Z19B** (\$20) - NDIR CO2 sensor (accurate, not estimated)
- **PMS5003** (\$25) - Laser PM2.5 sensor
- **SGP40** (\$10) - VOC sensor (10 sec warm-up)
- **AHT20** (\$5) - Temperature/humidity
- **OLED Display** (\$8) - Real-time feedback
- **Smart Plug** (\$12) - Purifier control

Hardware ordered: Week 1

Software Stack

Firmware: Arduino/PlatformIO (C++)

- UART protocol (CO2, PM2.5)
- I2C protocol (VOC, temp, display)
- MQTT communication

Backend: Firebase or Node.js + Express

Frontend: React (web dashboard)

Mobile: Progressive Web App initially, React Native later

ML: Python and its libraries

Project Features - Story Points Breakdown

Total: 100 Story Points

Core Features (60 points):

- Multi-sensor integration: 15 pts
- Real-time display: 5 pts
- Web dashboard: 10 pts
- Smart plug control: 10 pts
- Data logging: 10 pts
- Alert system: 10 pts

Project Features - Story Points Breakdown

Enhanced Features (30 points):

- Mobile PWA: 10 pts
- Data visualization: 8 pts
- Sleep quality tracking: 7 pts
- Cloud sync: 5 pts

Advanced Features (10 points):

- ML sleep prediction: 7 pts
- Multi-room support: 3 pts

Burndown Chart

Story Points Remaining Over Time

- **Week 4 (PPP):** 70 points remaining (30 points completed)
 - MVP: Sensors working, basic display
- **Week 8 (Mid-semester):** 35 points remaining (65 points completed)
 - Smart control, data collection running
- **Week 12:** 10 points remaining (90 points completed)
 - ML model trained, features complete
- **Week 14 (Final):** 0 points remaining (100% complete)

Success Metrics

Technical Metrics

- Sensor accuracy within $\pm 20\%$ of EPA reference monitors
- System uptime $> 99\%$ over 4 weeks
- Dashboard response time $< 500\text{ms}$
- Purifier control latency < 2 seconds

Project Metrics

- 4+ weeks of continuous data collected
- Working mobile app with historical data
- ML model with $> 70\%$ accuracy
- Automated purifier control functional

Learning with AI - Topic 1

Software Domain: Mobile App Development (React Native)

Why: Need cross-platform mobile app for remote monitoring and alerts

What I'll Learn:

- React Native framework and components
- Navigation and state management
- Push notifications (Firebase Cloud Messaging)
- Real-time data visualization
- API integration and async operations

How with AI:

- Claude for concept explanations and debugging

Learning with AI - Topic 2

Hardware Domain: Multi-Sensor Integration & Calibration

Why: 5 sensors with different protocols need accurate calibration

What I'll Learn:

- UART protocol (MH-Z19B, PMS5003)
- I2C bus architecture (SGP40, AHT20, OLED)
- Sensor datasheet interpretation
- Calibration techniques and signal processing
- Kalman filters and noise reduction

How with AI:

- Datasheet analysis and timing diagram explanation

Managing Three Work Domains

Tracking Progress in Each Area

Hardware Work (Sensor Integration):

- Weekly: Sensor bring-up, accuracy testing, calibration progress

Software Work (Web/Mobile Development):

- Weekly: Features completed, bugs fixed, UI improvements

ML Work (Data Analysis):

- Weekly: Data collected, models trained, insights discovered

Deliverables: Weekly progress reports for HW3 and final evaluation

Phase 1: Foundation (Weeks 1-4)

Sprint to PPP Demo

Week 1-2: Project Setup

- [DONE] HW2 requirements
- [DONE] GitHub repositories
- [DONE] Architecture design
- [IN PROGRESS] Hardware ordering

Week 3: Individual Sensors Working

- Test each sensor independently
- Document sensor behaviors
- Create sensor drivers

Phase 2: Smart Control (Weeks 5-8)

Data Collection & Automation

Week 5: Smart Plug Integration

- MQTT communication
- Automated purifier control
- Basic automation logic

Week 6: Data Logging

- Local and cloud storage
- Historical data access
- Data persistence testing

Week 7-8: Alerts & Visualization

Phase 3: Intelligence (Weeks 9-12)

Mobile App & Machine Learning

Week 9: Mobile App

- Progressive Web App
- Real-time updates
- Responsive design

Week 10: Multi-Room (Optional)

- Test with multiple units
- Room-to-room analysis

Week 11: Machine Learning

- Export 4+ weeks of data

Phase 4: Finalization (Weeks 13-14)

Documentation & Presentation

Week 13: Documentation

- Complete technical docs
- Installation guide
- User manual
- Health insights report

Week 14: Final Presentation

- Demo video
- Results compilation
- Project showcase

Week 4 Demo Plan

What I'll Show at PPP

Hardware Demo:

- ESP32 with all 5 sensors connected
- OLED display showing live readings
- Sensor placement at breathing zone height

Software Demo:

- Web dashboard accessible on laptop
- Real-time data updates
- Graphs showing last 24 hours

Control Demo (if smart plug arrives):

Risk Management

Identified Risks & Mitigation

Risk 1: Hardware Delays (Medium probability, High impact)

- Mitigation: Order immediately, develop software in parallel

Risk 2: Sensor Accuracy (High probability, Medium impact)

- Mitigation: Budget extra time for calibration, accept $\pm 20\%$ as limitation

Risk 3: MQTT Reliability (Medium probability, Medium impact)

- Mitigation: Local fallback, auto-reconnect logic

Risk 4: Time Constraints (High probability, Medium impact)

- Mitigation: Prioritize core features, make ML optional

Current Progress (Week 1)

Completed This Week

- Project planning and research
- GitHub repositories created:
 - smart-home-health-monitor
 - mausam-shrestha-learning-with-ai
- Comprehensive README files
- 14-week project plan with milestones
- Canvas pages created
- PPP presentation prepared

Next Immediate Steps

Expected Impact

Technical Contributions

- Demonstrates location bias problem with commercial purifiers
- Open-source air quality monitoring solution
- Validates AutoML calibration for low-cost sensors
- Correlates air quality with sleep outcomes

Personal Growth

- Full-stack IoT development experience
- Mobile app development skills
- ML/data science application
- Portfolio piece for job applications

Why This Project Will Succeed

Strong Foundation

1. **Clear Problem:** Research-validated pain points
2. **Achievable Scope:** MVP in 4 weeks, features incrementally
3. **Market Validation:** \$200-400 commercial products prove demand
4. **Risk Management:** Mitigation strategies for each risk
5. **AI-Assisted Learning:** Using Claude for efficient skill acquisition
6. **Realistic Timeline:** 10-12 hours/week, 140-168 total hours

Repository Links

Project Code:

<https://github.com/shresthamausam07/smart-home-health-monitor>

Learning Documentation:

<https://github.com/shresthamausam07/mausam-shrestha-learning-with-ai>

Canvas Project Page:

<https://nku.instructure.com/courses/88152/pages/individual-project-mausam-shrestha>

Canvas Progress Page:

<https://nku.instructure.com/courses/88152/pages/individual-progress-mausam-shrestha>

Questions & Discussion

Ready to Answer Questions About:

- Technical approach and architecture
- Sensor selection rationale
- Timeline feasibility
- Risk mitigation strategies
- Learning topics and AI usage
- Success metrics and validation
- Scalability and future enhancements

Thank you!

